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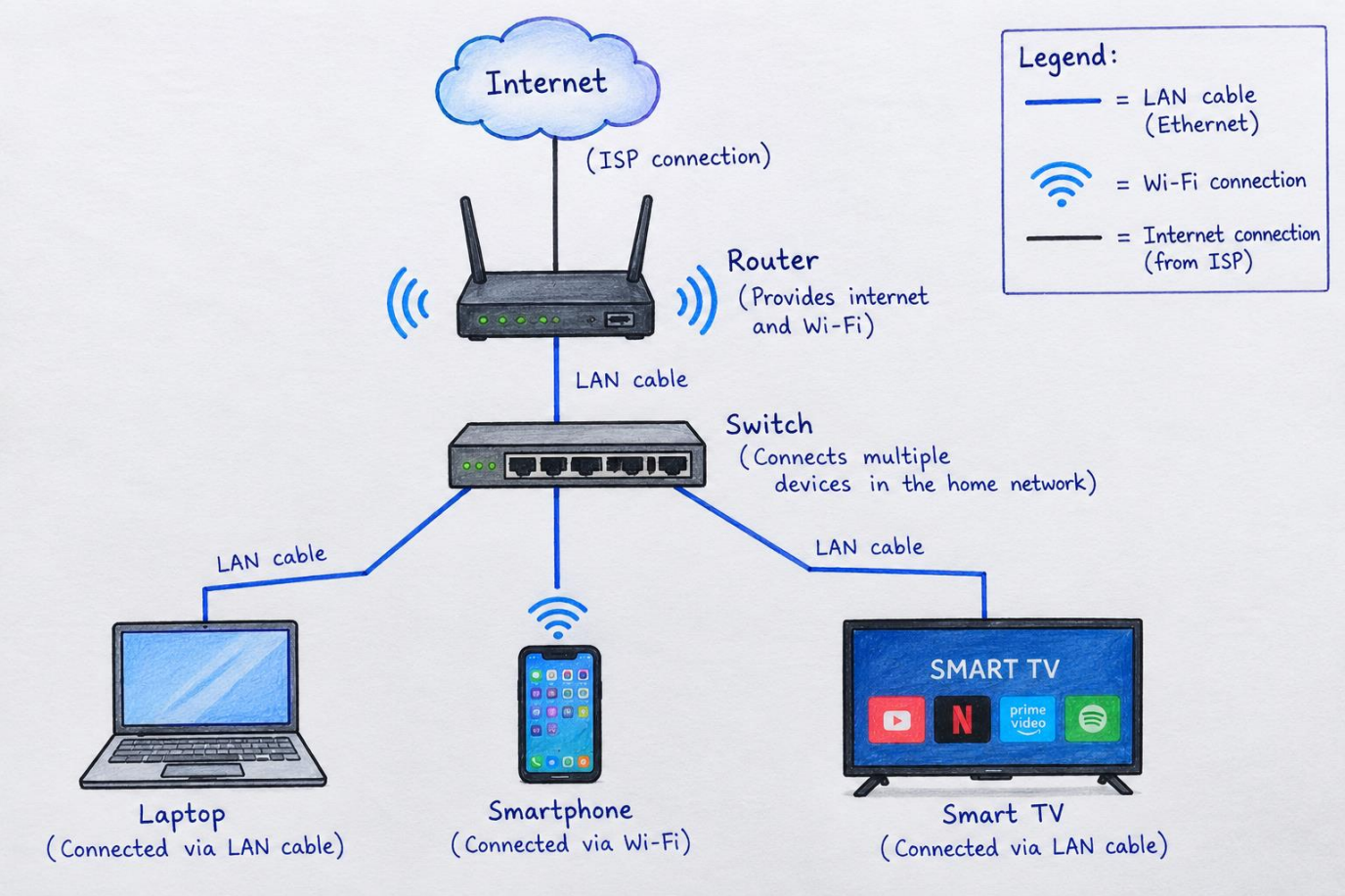
N+ Network Fundamentals and Buillding Networks

Network devices

TASKS:-1

Draw and label a simple home network diagram showing a router, switch, and three devices (like a laptop, smartphone, and smart TV), indicating how each device connects to the network.

RESULT



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TASKS:-2

Create a comparison table listing at least four network devices (router, switch, hub, access point) and describe one real-world use case for each, referencing apps or services you use daily (e.g., WiFi for streaming on YouTube, LAN gaming, etc.)

RESULT

Network Device	Main Function	Real-World Use Case
Router	Connects different networks and provides Internet access.	I use a Wi-Fi router to connect my phone and laptop to the Internet for YouTube, WhatsApp, and browsing.
Switch	Connects multiple devices within a LAN and forwards data to the correct device.	A switch connects PCs in a computer lab for LAN gaming and file sharing.
Hub	Connects multiple devices and broadcasts data to all connected devices.	A hub can connect old computers in a small lab for basic network communication.
Access Point	Provides wireless Wi-Fi connectivity to devices.	An access point connects smartphones and Smart TVs to Wi-Fi for Netflix, YouTube, and online streaming.

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TASKS:-3

Write a short scenario where you explain how data travels from your phone to the internet when you order food on Zomato, mentioning the role of each network device involved (router, switch, access point).

RESULT

Scenario: Ordering Food on Zomato

When I order food using the Zomato app on my smartphone, my phone connects to the home Wi-Fi through the Access Point. The data travels to the Router, which forwards the request to the Internet through my ISP. Switches help transfer data between network devices in the network. The request reaches Zomato's server, where my order is processed. Finally, the confirmation and delivery details are sent back to my phone through the Internet.

Role of Network Devices:

1. **Access Point**: Connects my smartphone to Wi-Fi.
2. **Router:** Connects my home network to the Internet.
3. **Switch**: Transfers data between connected devices in a network.

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TASKS:-4

Given this situation: you have a WiFi router, but your gaming PC only has an Ethernet port and is far from the router. Suggest two different network devices you could use to connect your PC to the internet and explain how each would work in this setup. Think about devices like switches, powerline adapters, or WiFi extenders

RESULT

Connecting a Gaming PC to Wi-Fi Router

If my gaming PC has only an Ethernet port and is far from the Wi-Fi router, I can use the following two network devices:

1. Powerline Adapter

A powerline adapter uses the existing electrical wiring of my home to transfer network data. I connect one adapter to the router using an Ethernet cable and plug it into a power socket. The second adapter is plugged near my gaming PC and connected using another Ethernet cable. This provides a wired Internet connection for gaming.

2. Wi-Fi Extender with Ethernet Port

A Wi-Fi extender receives the wireless signal from the router and extends it to a distant area. I can place it near my gaming PC and connect the PC to its Ethernet port using a LAN cable. This allows my gaming PC to access the Internet without running a long cable from the router.